

Saber Ayatollahi, CFA, MSc

Toronto, Canada • +1 (416) 856-1276 • saber.ayatollahi@gmail.com • www.linkedin.com/in/sayatollahi

SUMMARY

Senior Manager, Risk Analytics & Modelling candidate with ~7 years across instrument risk modelling, derivatives valuation, and multi-asset investment risk analytics for institutional portfolios. Delegated sign-off authority at Moody's Analytics covering rates, FX, and inflation derivatives; hands-on VaR/CVaR portfolio optimization, risk decomposition, and stochastic scenario design at Ortec Finance for Canadian pension funds. Strong Python and vendor risk-system depth; agentic AI workflows with LLMs in production. CFA charterholder, dual MSc (Financial Modelling; Chemical Engineering).

CORE SKILLS

Instrument Risk Modelling & Derivatives Valuation • Investment & Market Risk Analytics • VaR / CVaR & Risk Decomposition • Stochastic Modelling & Monte Carlo Simulation • Model Validation, Benchmarking & Documentation • Pension Liability & ALM Modelling • Python (pandas, NumPy, SciPy) & SQL • Vendor Risk Systems (PFaroe, Ortec GLASS) • AI Agents & LLM-Powered Automation

EXPERIENCE

Moody's Analytics

Asst. Director, Modelling Services | Toronto, May 2023 – Present

Instrument Risk Modelling & Derivatives Valuation

- Hold delegated sign-off authority on valuation, sensitivity, and risk outputs for multi-asset institutional portfolios ranging \$5-25bn per engagement, cumulatively covering ~\$50bn of client assets.
- Validate derivatives pricing outputs across rates, FX, and inflation; cross-check sensitivities and scenario impacts at portfolio-level aggregates for institutional clients including pension funds, asset managers, and their portfolio managers.
- Review curve construction, spread calibration, and cross-asset interactions prior to production release; escalate outputs lacking mathematical or economic defensibility.

Market Risk Tools, Scenario Analysis & Risk Reporting

- Led design and implementation of an enterprise multi-asset cash-flow projection engine supporting base, stress, and reverse-stress scenarios across portfolio horizons.
- Architected configurable time-bucketed exposure and liquidity analytics (T+1 through multi-year) supporting hedging, funding, and asset-allocation decisions.
- Oversaw interest-rate and duration analysis under parallel and non-parallel shocks; prepared analytical commentary translating quantitative outputs into investment-committee-ready risk reports.

Model Validation, Python & AI-Powered Automation

- Operate within a formal model governance framework – independent review and challenge, documentation, benchmarking, and assessment of model performance for client-delivered analytics.
- Re-engineered manual spreadsheet workflows into scalable, auditable Python analytics pipelines with embedded logging and validation controls.
- Built agentic AI development workflows leveraging LLMs (Claude Code, Cursor IDE) – automating code generation, validation scaffolding, anomaly detection, and documentation; reduced cycle time ~30-40% on comparable modules.

Client Service Specialist | Toronto, May 2022 – May 2023

- Led onboarding and implementation of the PFaroe vendor risk and analytics platform for U.S. and Canadian pension funds, asset managers, and consulting firms.
- Served as technical liaison between client investment teams and product/development functions – scoped requirements, advised on asset, liability, and ALM configurations, and validated model outputs post-deployment.
- Contributed to the successful migration of all assigned client accounts from a legacy Calypso environment to the PFaroe platform.

Ernst & Young (EY)

Senior Consultant, IFRS & Financial Risk Transformation | Toronto, Sep 2021 – Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients – aligned liability measurement and reporting frameworks with system implementation.
- Advised actuarial, finance, IT, and risk functions on governance documentation, model controls, and regulatory readiness across complex insurer balance sheets.

Ortec Finance Canada

Consultant, Asset-Liability Management & Risk | Toronto, Feb 2019 – Sep 2021

- Delivered ALM studies for Canadian and international pension funds – analyzed funding ratios, duration mismatch, and long-term balance-sheet risk, modelling actuarial liabilities alongside all major investment-product types and economic variables.
- Performed asset-only and asset-liability (surplus) portfolio optimization based on VaR and CVaR using Ortec's GLASS vendor risk system; conducted risk decomposition, contribution-to-risk, and near-optimal frontier robustness analysis.
- Built and interpreted stochastic economic scenario generators assessing interest-rate, inflation, and currency risk under base and stressed environments; performed hedging analysis across rates, inflation, and FX.
- Conducted liability-driven investment (LDI) analysis trading off return objectives, duration alignment, and liquidity; presented findings to investment committees on Strategic and Tactical Asset Allocation through stress and sensitivity testing.
- Led onboarding and the ALM study for the University Pension Plan (UPP) three-plan merger – on a team of three, developed and validated the model capturing funding-policy dynamics, evaluating duration, currency, inflation, and leverage overlay strategies.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) – CFA Institute, 2024
- MSc, Financial Modelling – Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) – Western University, 2016